Advanced Database Management Systems Lab
Lab Task 9

[CLO-2, C3, GA5] **[20 Marks]**

Submission

- Submit SQL scripts
- Include screenshots of outputs
- Provide explanation of each step
- Highlight any consistency issues observed and how they were resolved

StreamFlix is a global video streaming platform operating in multiple regions including, Islamabad Region, Lahore Region, and Karachi Region. Each region manages its own users, viewing activity and content streaming. The Head Office requires centralized analytics, user behavior reports, and system-wide monitoring. Due to large-scale data, the data is distributed across regions, replication is used for analytics and consistency must be maintained.

.Schema

Table 1: Regions	**Table 2: Users**
CREATE TABLE regions (region_id SERIAL PRIMARY KEY, region_name VARCHAR(100), city VARCHAR(50));	CREATE TABLE users (user_id SERIAL PRIMARY KEY, user_name VARCHAR(100), email VARCHAR(100), subscription_type VARCHAR(50), region_id INT REFERENCES regions(region_id));
Table 3: Content	
CREATE TABLE content (content_id SERIAL PRIMARY KEY, title VARCHAR(100), genre VARCHAR(50), duration INT, release_year INT, content_rating VARCHAR(10));	
Table 4: Streaming Activity	
CREATE TABLE streaming_activity (activity_id SERIAL PRIMARY KEY, user_id INT REFERENCES users(user_id), content_id INT REFERENCES content(content_id), region VARCHAR(50), watch_time INT, device_type VARCHAR(50), activity_date DATE);	

Tasks

1. Create the following base tables using the provided schema:
    - regions
    - users
    - content
    - streaming_activity

    Insert appropriate sample data:
    - At least 3 regions (Islamabad, Lahore, Karachi)
    - At least 5 users linked to regions
    - At least 5 content items
    - At least 6-8 streaming activity records
2. **Horizontal Fragmentation of Streaming Activity**

Create separate tables based on region:
- streaming_islamabad
- streaming_lahore
- streaming_karachi

Fragment the streaming_activity table using the condition:
- region = 'Islamabad'
- region = 'Lahore'
- region = 'Karachi'

3. **Vertical Fragmentation of Content Table**

Split the content table into two tables:

Table A:

Columns:
- content_id
- title
- genre

Table B:

Columns:
- content_id
- duration
- release_year
- content_rating

4. **Reconstruction of Vertically Fragmented Data**

Reconstruct the original content table by performing a JOIN operation between both fragments

5. **Replication for Centralized Monitoring**

Create a replica of the streaming_activity table: Name: streaming_replica_hq
This table represents the Head Office copy

6. **Distributed Queries**
Perform the following queries using fragmented tables:
    - Calculate total watch time across all regions
    - Generate region-wise watch time report
    - Identify the most watched content
7. **Insert Operation and Fragment Update**
Insert a new record into the main streaming_activity table and identify which fragment it belongs to. Also, insert the same record into the appropriate fragment table
8. **Update Operation and Consistency Check**
Update the watch_time of a record in one fragment table
Compare the same record in Main table, Replica table and identify any inconsistency in values
9. **Consistency Issue Identification**
Explain:
    - Why inconsistency occurred
    - Which tables are out of sync
    - Provide a short explanation of:
    - Distributed update challenges
10. **Fixing Consistency**
Synchronize data by updating replica or main table accordingly and verify all tables now show consistent values
11. **Delete Operation and Inconsistency**
Delete a record from one fragment table and check whether the same record still exists in Main table, Replica table and identify resulting inconsistency
12. **Correcting Deletion Across System**
Remove the same record from Main table, Replica table and verify data consistency is restored across all tables